

Two-Part Lecture:

Broad overview of GIA and model components.
(~30 mins)

GIA modelling and model applications (~60 mins).

Outline

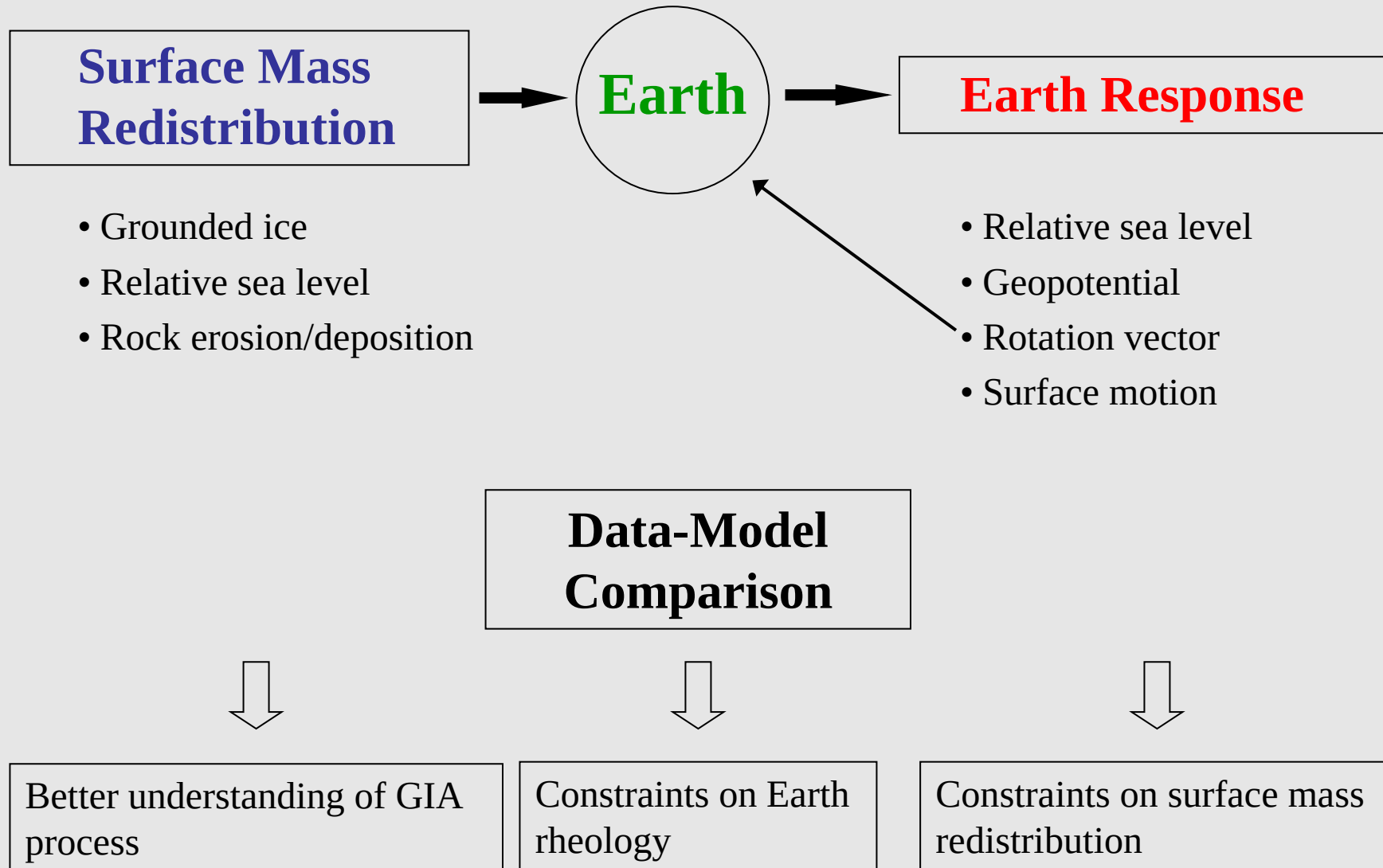
Model applications

Model tuning/calibration

Interpreting data-model misfit

Model accuracy/uncertainty

GLACIAL ISOSTATIC ADJUSTMENT



GIA Modelling

- (1) Near-field regional calibration to constrain regional Earth viscosity and ice sheet history.
- (2) Can also tune model parameters to other data sets (e.g., far-field, rotational data).
- (3) Use of calibrated models in a variety of applications (end users).

Focus here is (1).

GIA Model Applications

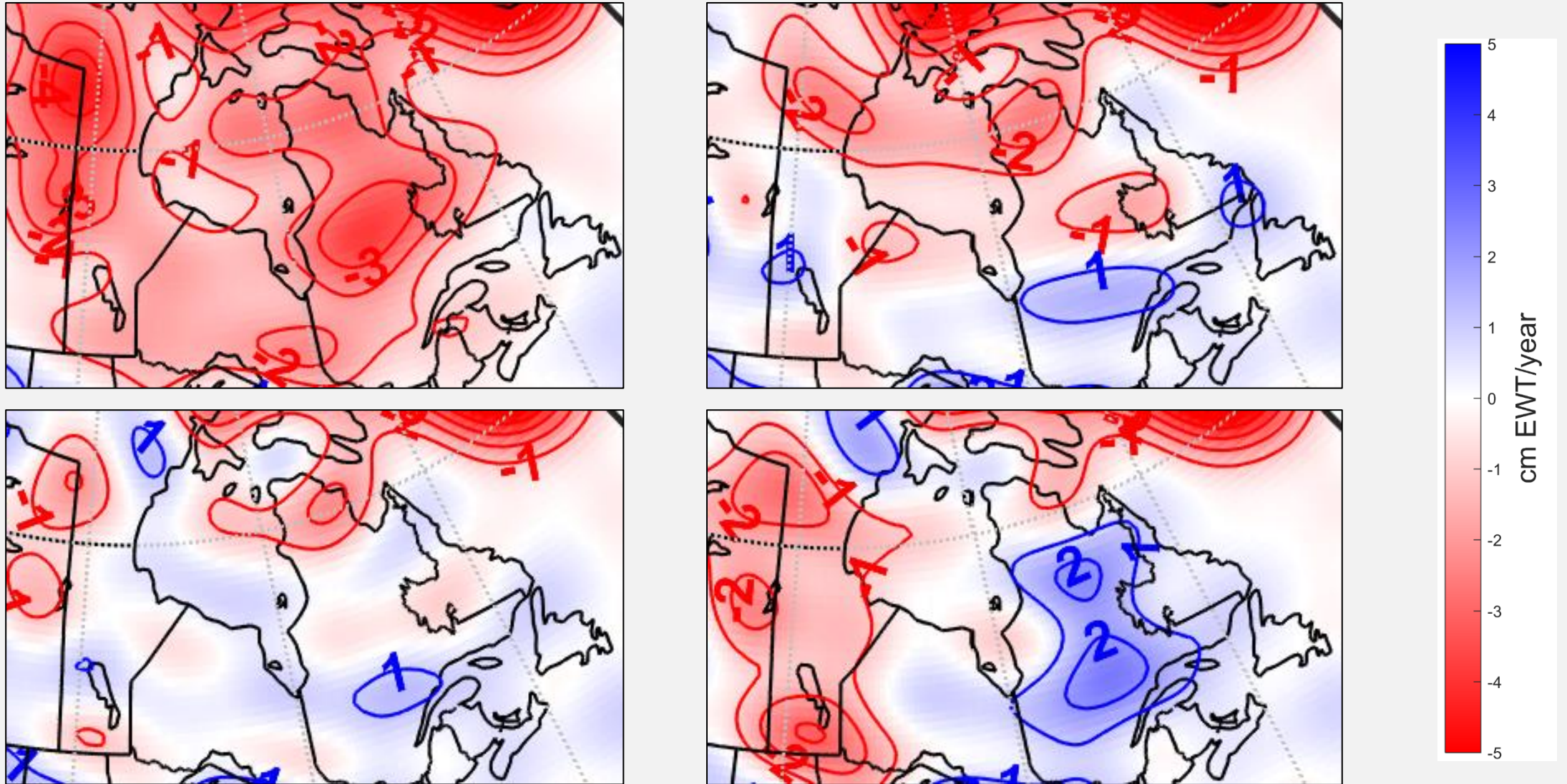
A calibrated GIA model has a number of applications.

A principal one is to remove the GIA contribution to a data set so the “cleaned” data can be used to make more accurate inferences of other quantities, for example:

(1) Inferring contemporary ice mass or hydrological changes from GRACE data; (2) Estimating parameters in geodynamic models of tectonic processes; (3) Estimating global ice volume from far-field sea-level observations.

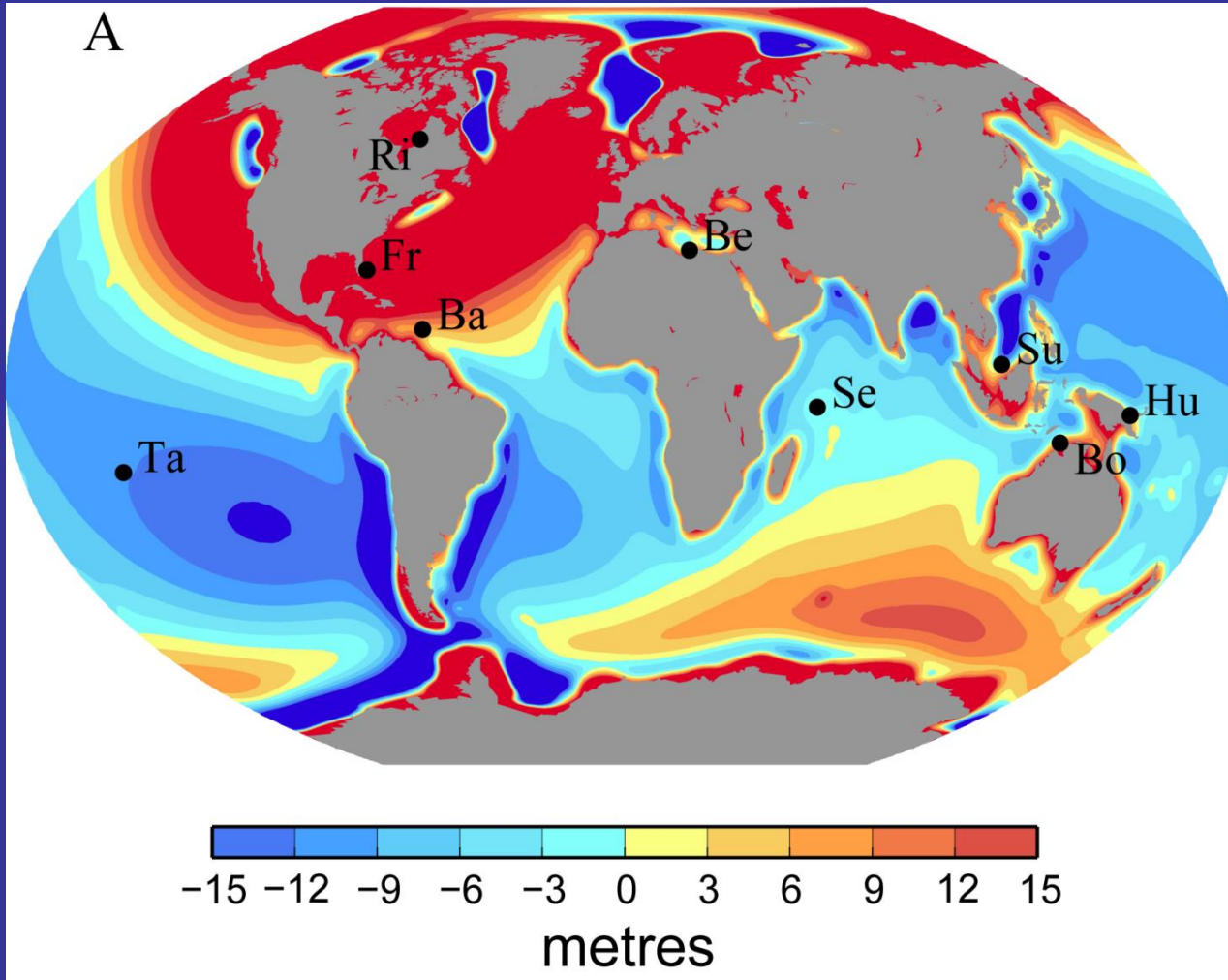
In other cases, we seek to add a number of component signals (including GIA) to get the total – e.g., sea-level projections.

GRACE Total Water Storage Trends



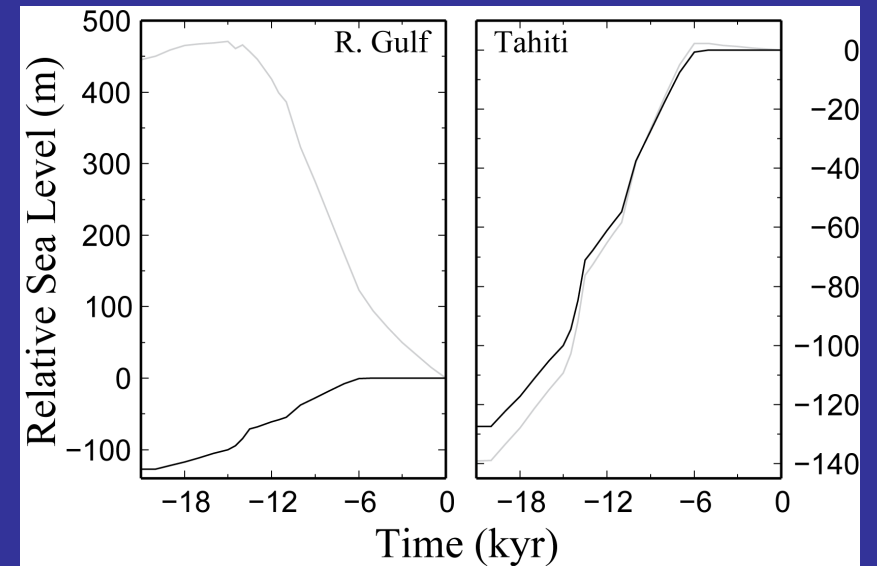
Courtesy of John Crowley, NRCan

Departures from Eustasy at the LGM

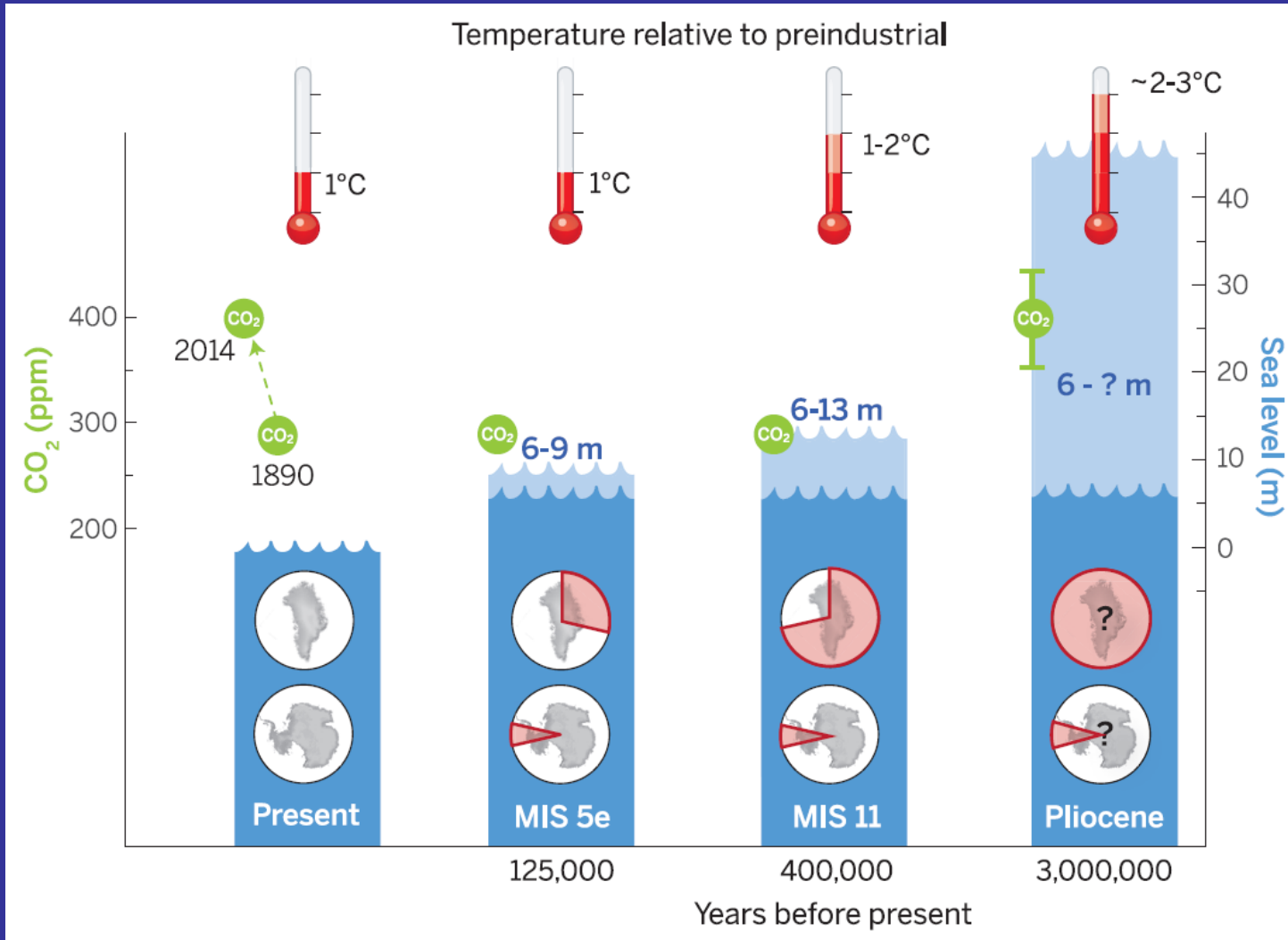


Milne & Mitrovica (2008)

Most local sea-level indicators will contain a GIA 'overprint' that should be removed to estimate past grounded ice volume.



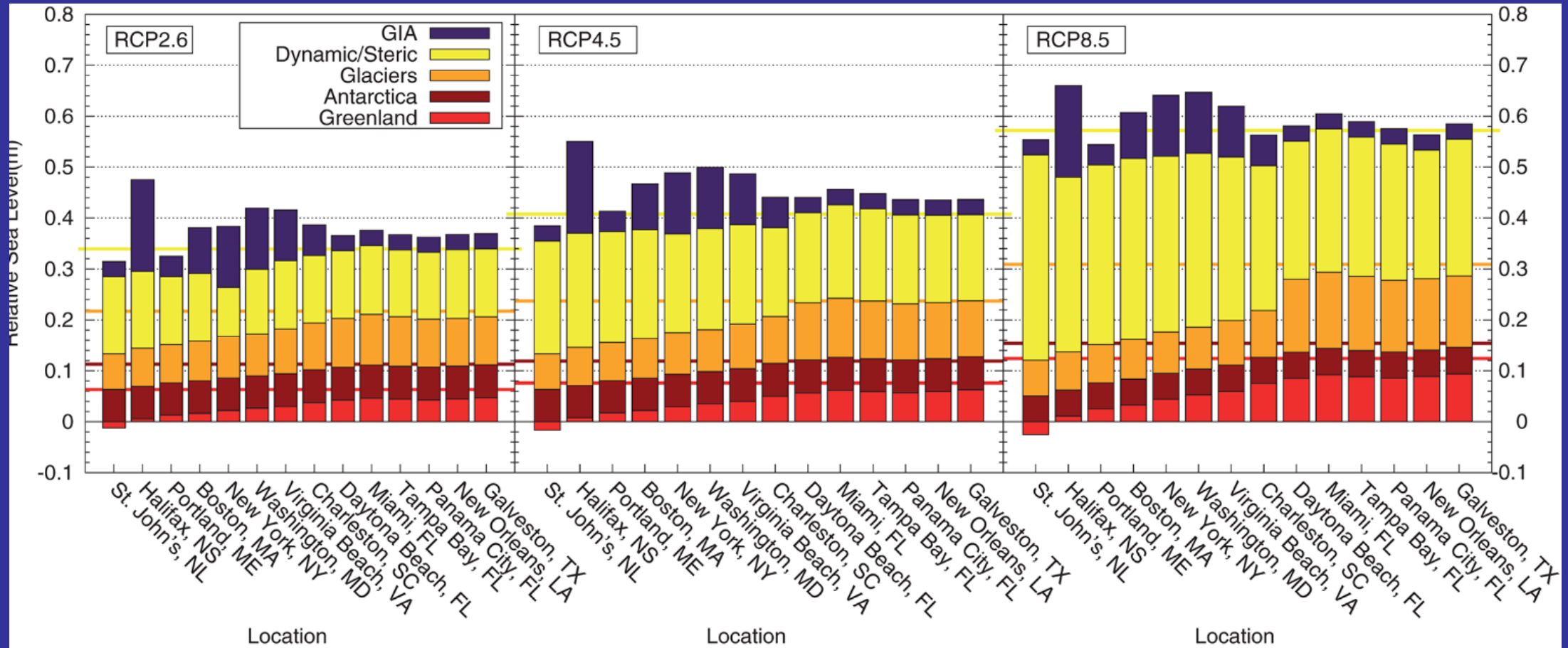
Paleo Analogues for GMSL



GIA is an important “correction” to apply when estimating global mean sea level from local sea-level reconstructions.

For older times (LIG onwards), sediment isostasy is also important.

Sea-Level Projections (I): GIA

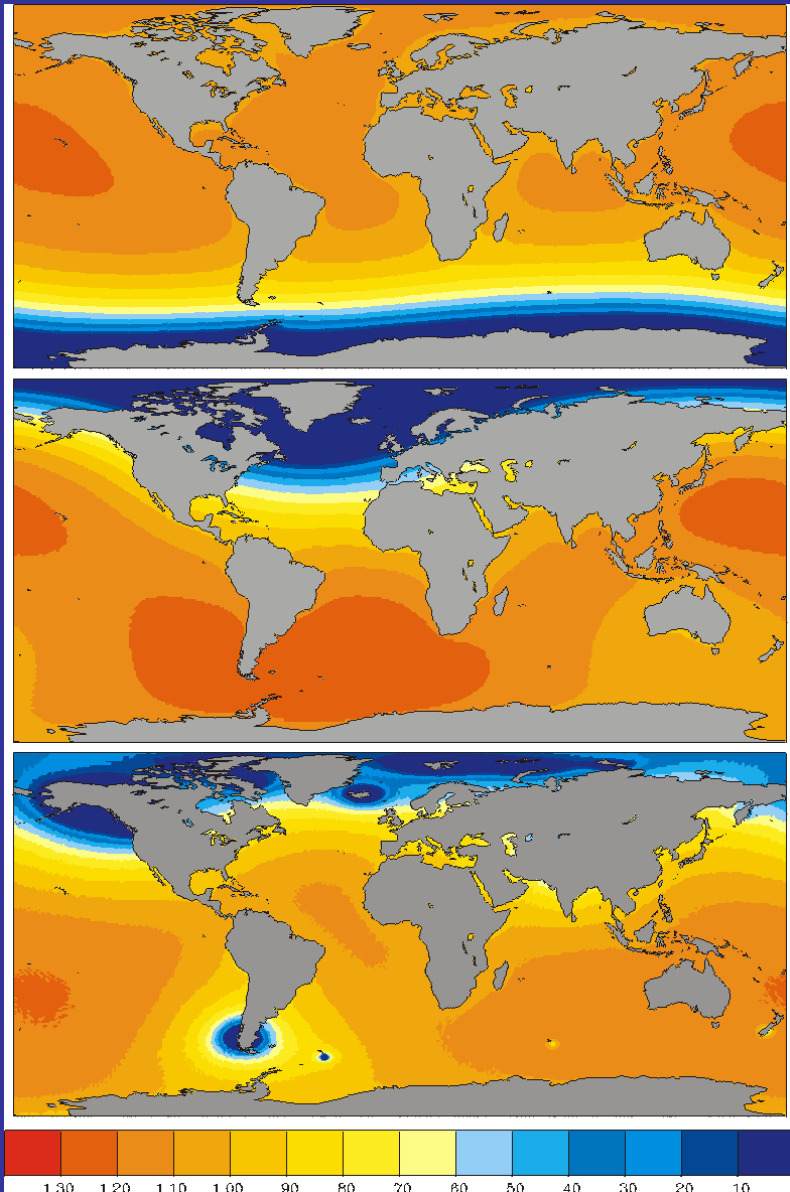


Love et al. (2016)

Sea-Level Projections (II): Fingerprints

Another key aspect of making sea-level projections that come from GIA models.

Since relatively short time scales are considered (~100 years), non-elastic component of response is small in most regions.



Mitrovica et al. (2001b)

GIA Modelling and Model Applications

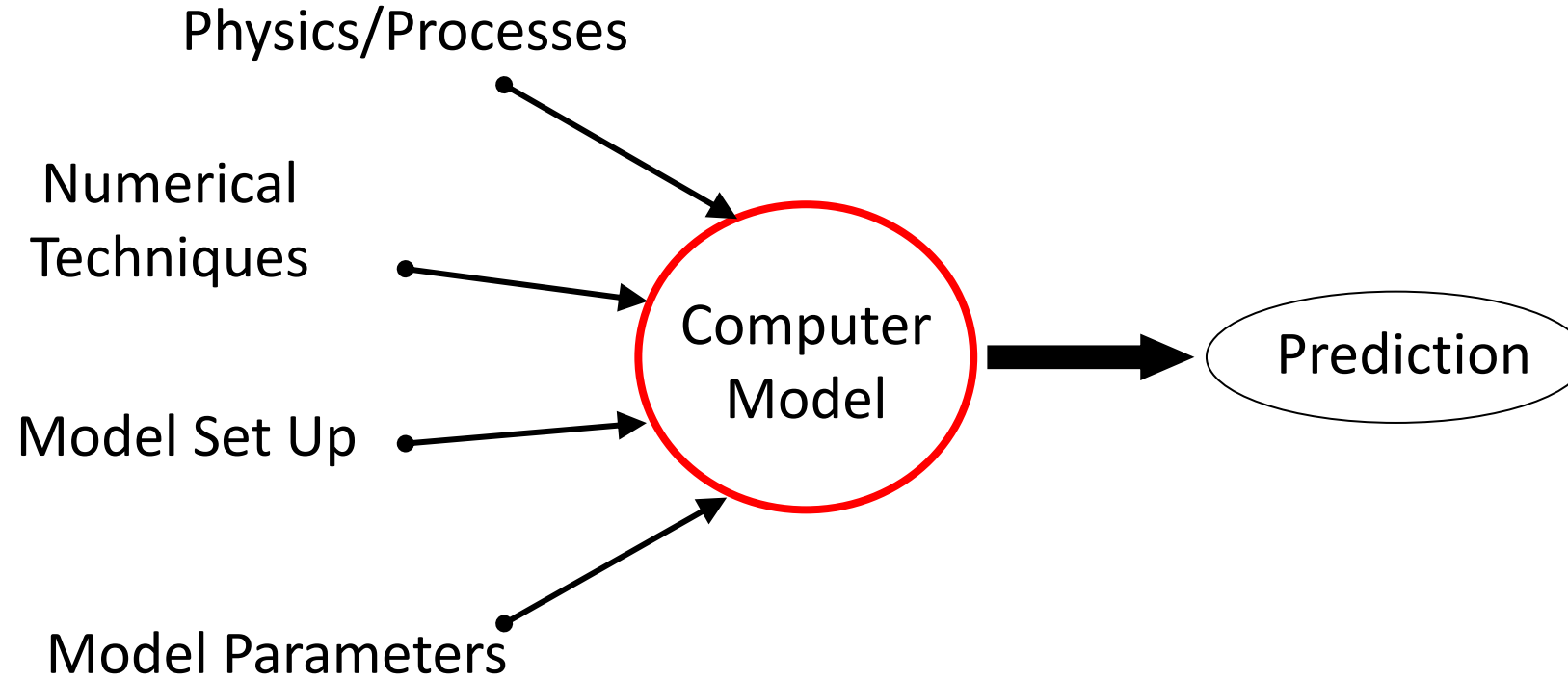
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General Elements of a Computer Model



Some Key Points

- Computer (numerical) models are simplified representations of the real world.
- Computer models are tools with which scientists can test hypotheses or make inferences by comparing model output to observations.
- Computer models represent our understanding of the natural world and can be used to extrapolate beyond observational coverage (in space and time).

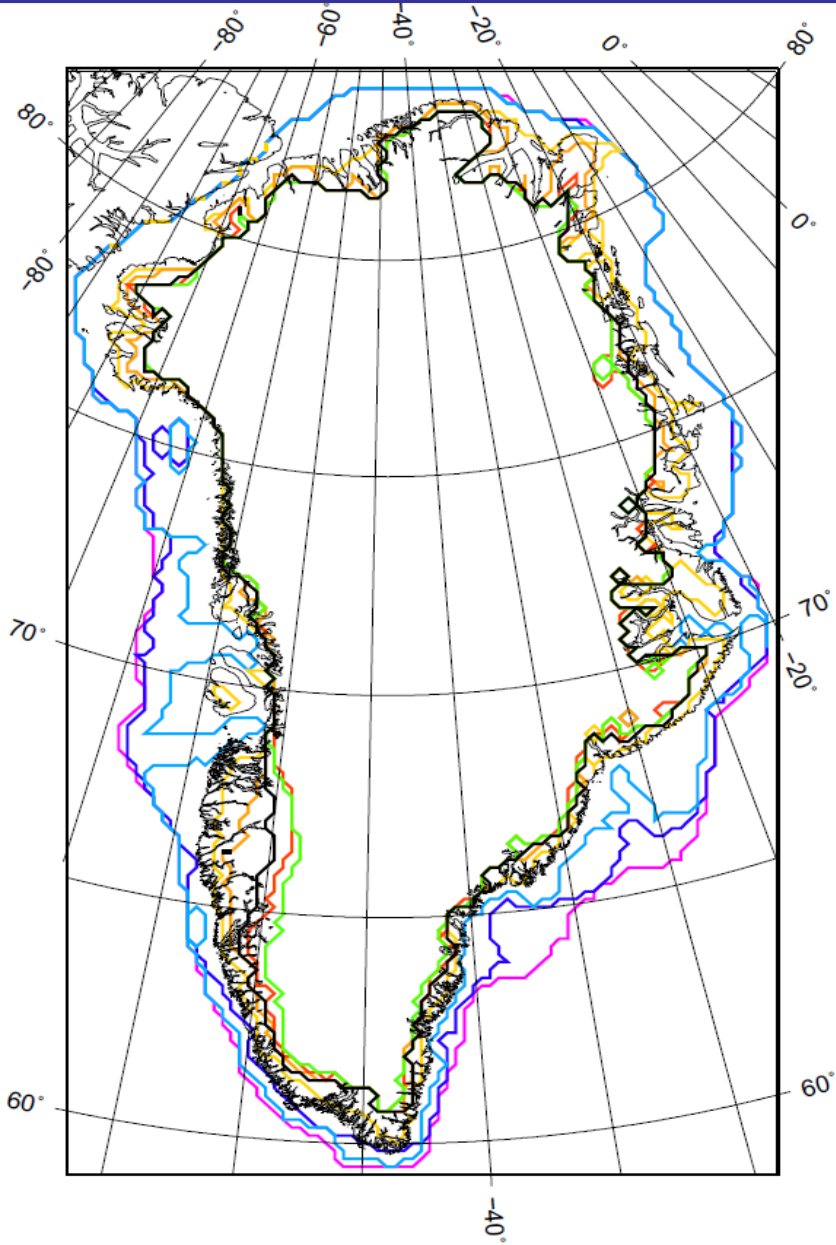
Overview of Parameter Inference Procedure

- Consider Greenland as a case study.
- Highlight similarities and differences with some other studies.
- Touch upon some central issues: solution non-uniqueness, model accuracy/uncertainty.

Parameter Inference Procedure: Forward vs Inverse

- Forward modelling involves producing model output for many runs (e.g., different parameter sets) and assessing model fit using a quantitative measure of the model (mis)fit. Statistical methods applied to determine model bounds or range of best-fitting model parameters.
- Inverse modelling involves the application of inverse theory (algorithms) to determine an optimal parameter set for a given forward model and data set. Requires fewer forward runs.
- Most GIA analyses have adopted forward modelling approach. For an example of the inverse approach, see Mitrovica & Forte (2004).

Step 2: Constrain Regional Ice Model Reconstruction

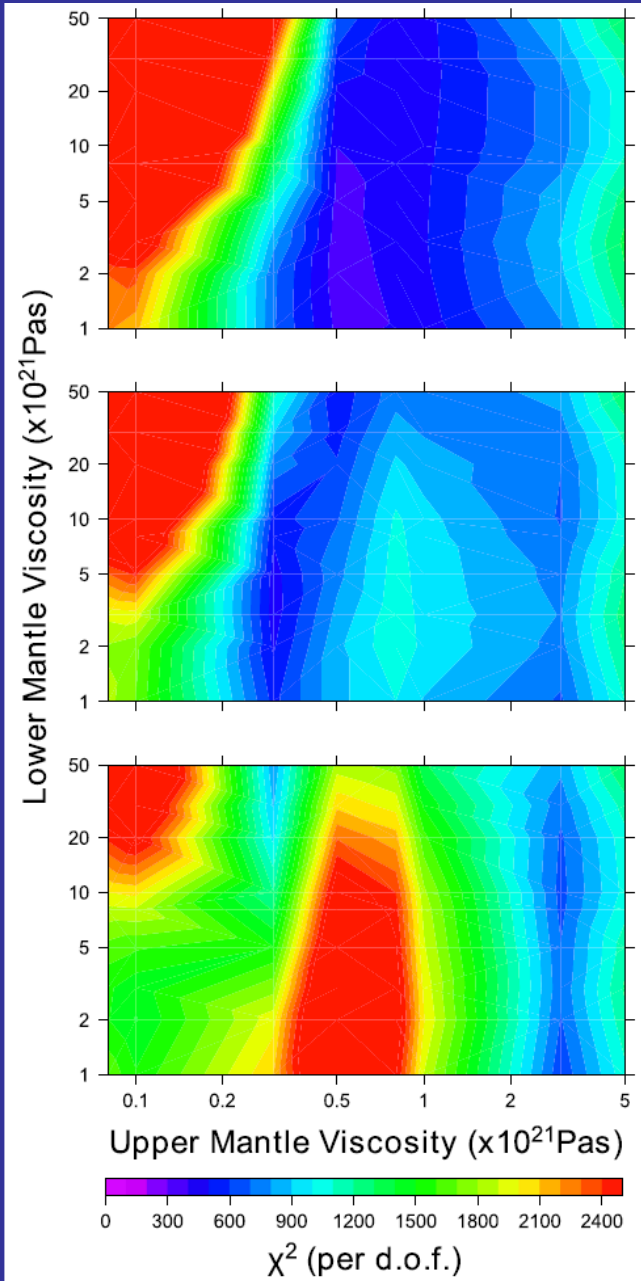


Vary LGM mask and sensitivity to sea-level forcing to match LGM extent and 'hit' any dated margin features after this time.

Looking for earlier retreat in east compared to west.

Lecavalier et al. (2014)

Step 3: Seek Optimal Earth Viscosity Parameters



Compute data-model RSL misfit for each set of Earth model parameters.

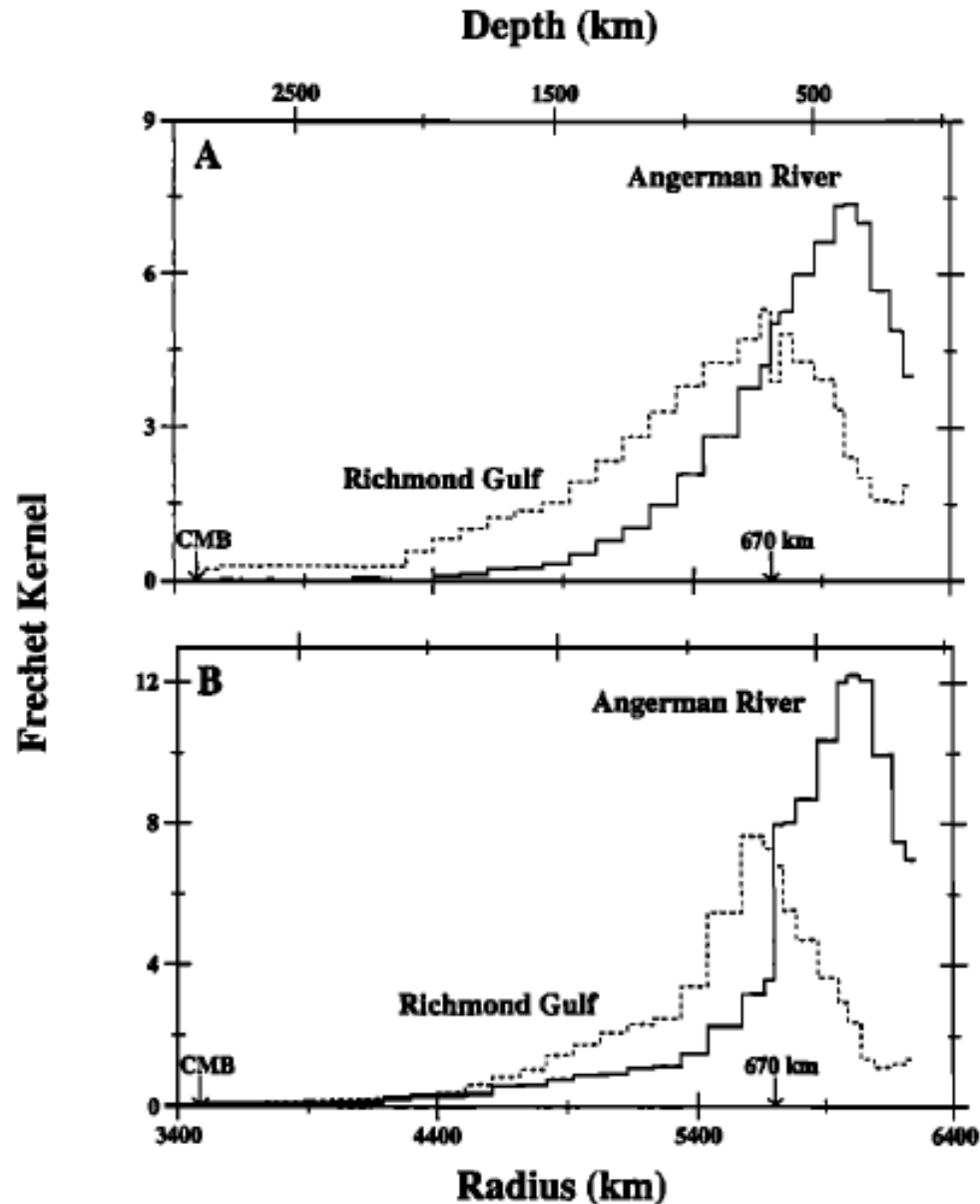
Optimal parameter set minimises the misfit parameter.

There are a number of models that can fit the data to a given statistical confidence.

Solution non-uniqueness.

Lecavalier et al. (2014)

Model Resolving Power



Model parameter sensitivity varies greatly.

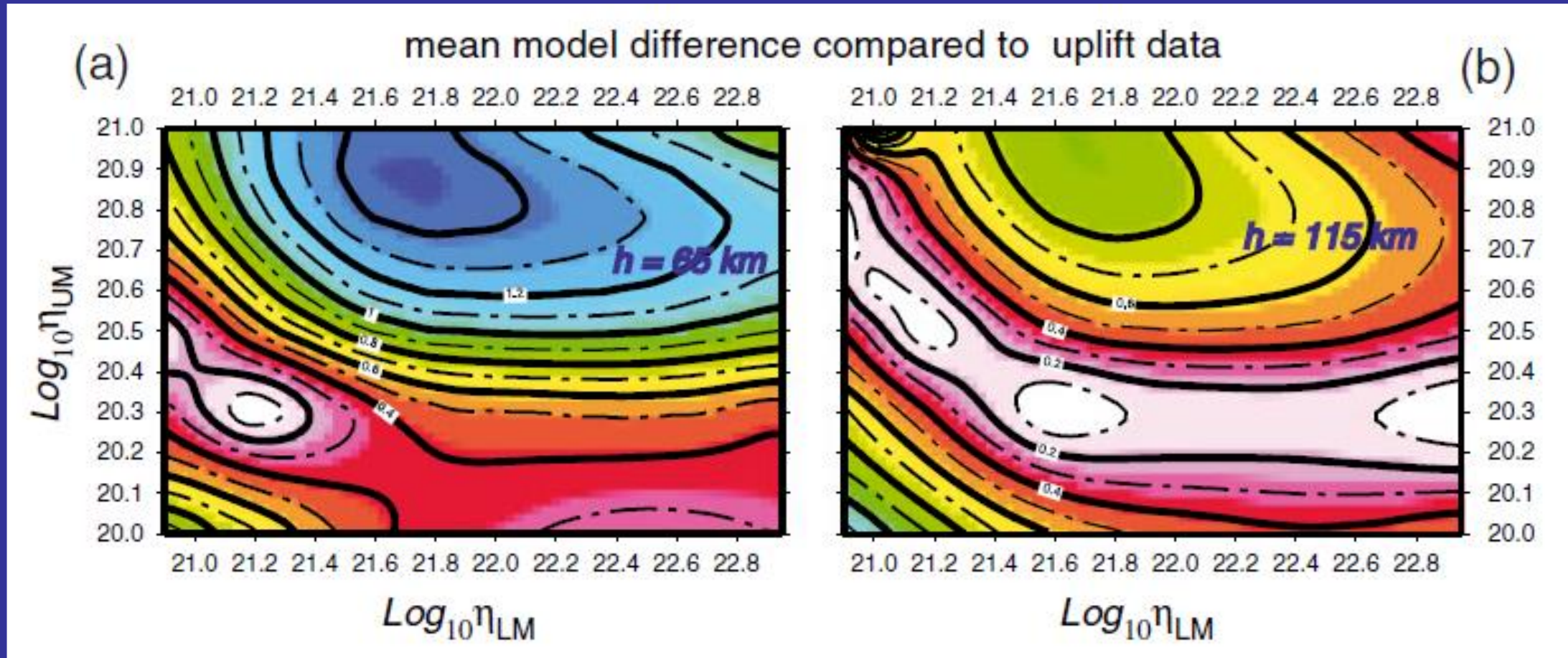
For a given problem it is important to realise which model parameters are poorly resolved.

This can be done via model sensitivity tests or more formally using inverse theory.

Other Approaches

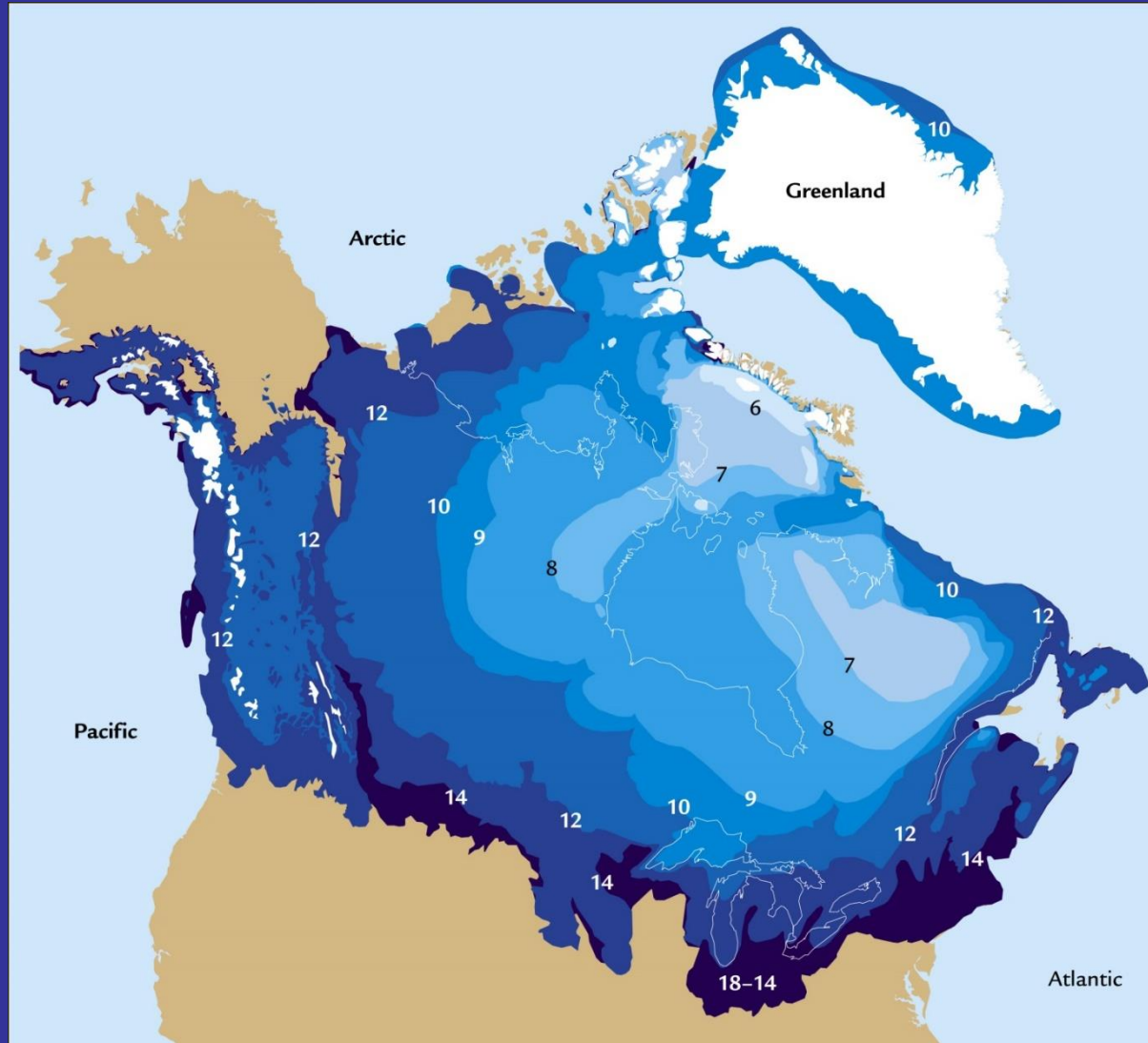
- Use relatively simplistic ice parameters and iteratively change ice and Earth model parameters in seeking optimal ice and Earth parameters (e.g., Lambeck et al., 1998).
- Assume that the Earth viscosity structure is known and vary ice thickness distribution until model fits GIA observations (e.g., Peltier, 2004).
- Perform a much more exhaustive search of ice model parameter space assuming a given Earth viscosity model and apply Bayesian statistics to determine model uncertainty (e.g., Tarasov et al., 2012).
- Make use of GPS observations when inferring optimal parameters (e.g., Whitehouse et al., 2012; Ivins et al. 2013).

Data-Model Misfit: Antarctic GPS Data



Ivins et al. (2013)

Reducing Degree of Non-Uniqueness (I)



Use data that are sensitive to only/mainly one aspect of the model.

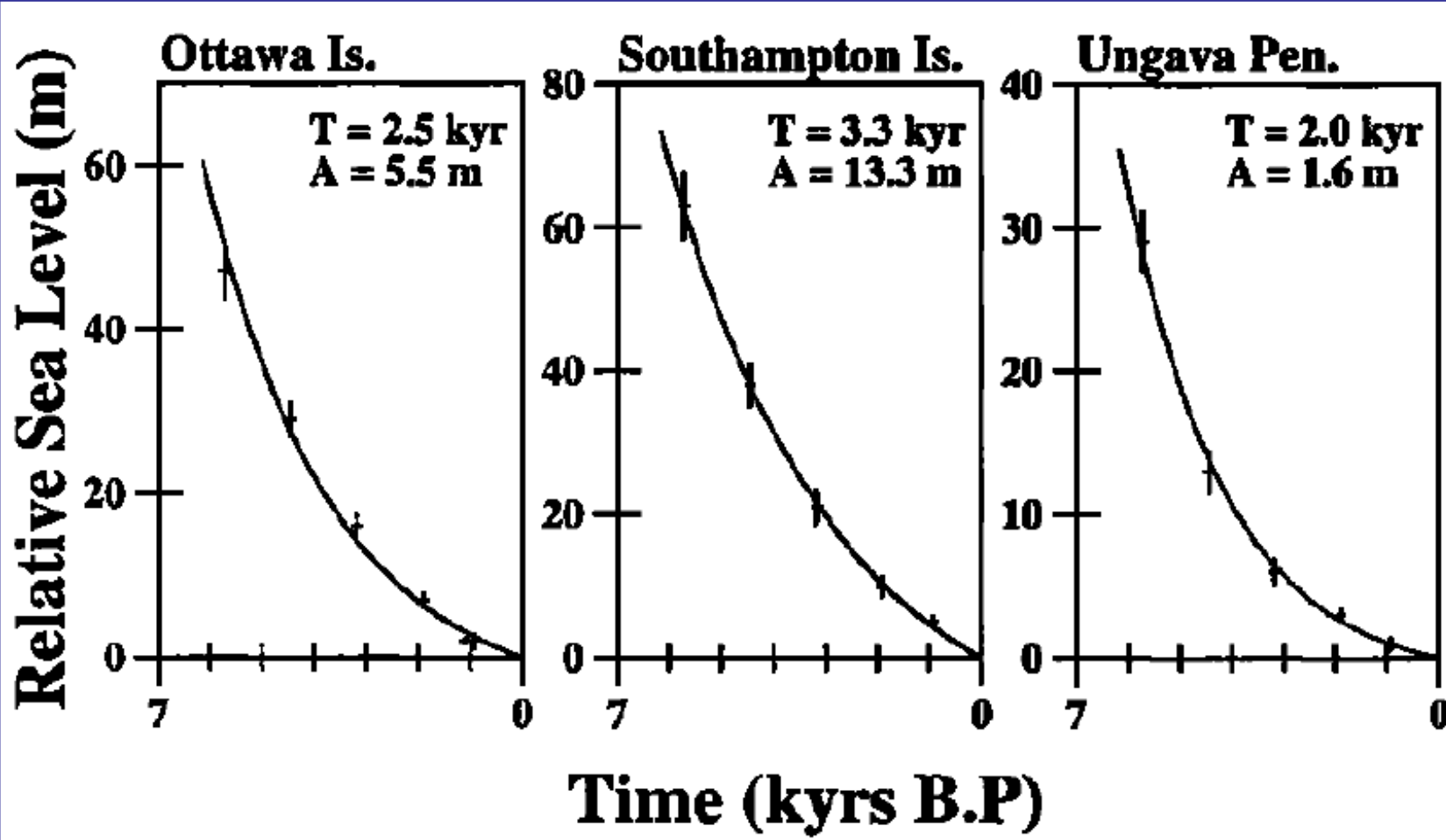
Ice extent data are key in this regard (but we need more!)

Reducing Degree of Non-Uniqueness (II)

Fit near-field observations with a simple exponential form:

$$RSL_i(t) = A_i \left[\exp\left(\frac{t}{\tau_i}\right) - 1 \right]$$

The so-called “decay time” (Tau) is relatively insensitive to the ice history.



Mitrovica & Peltier (1993), Kuchar et al. (2020)

GIA Modelling and Model Applications

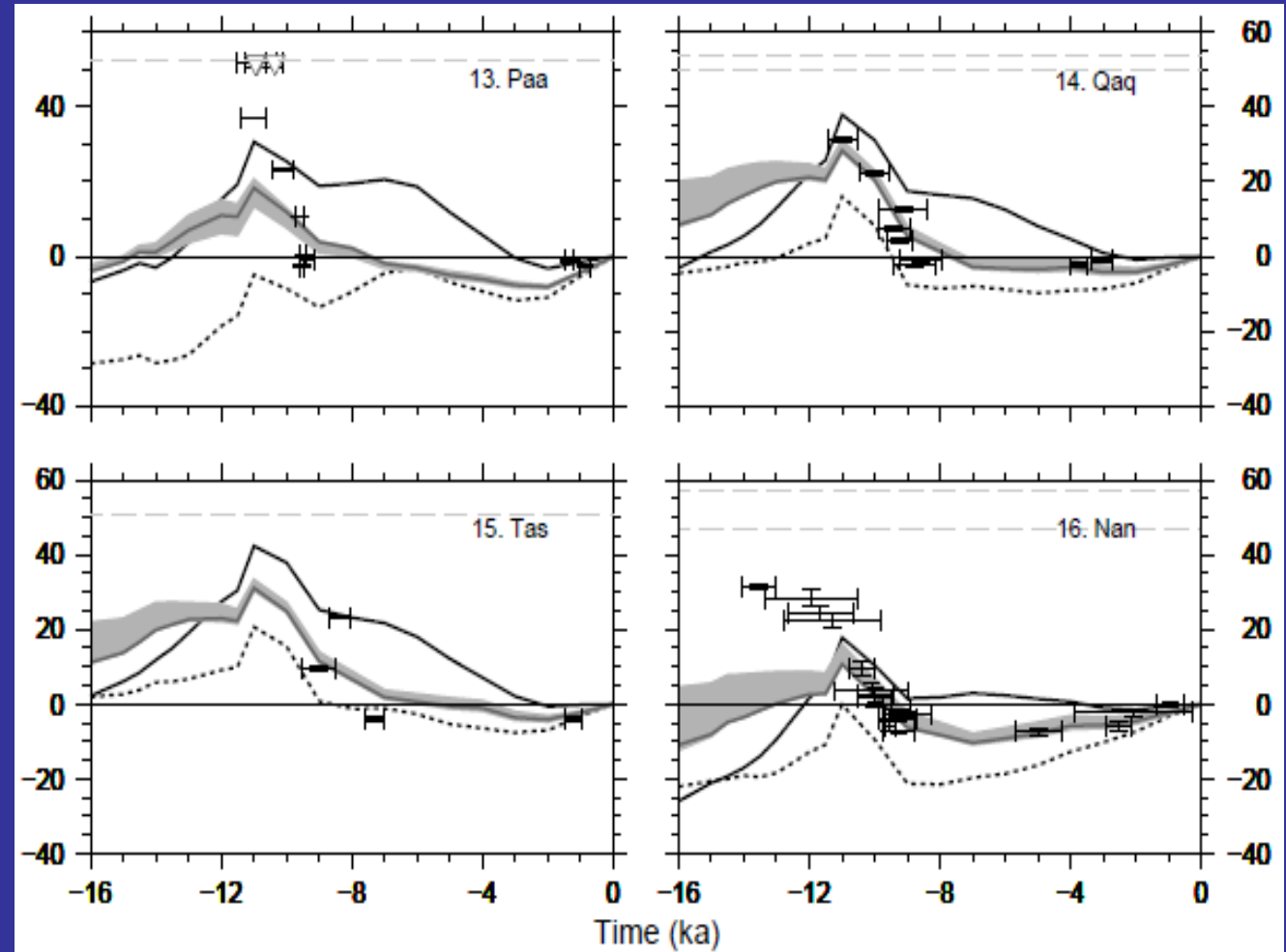
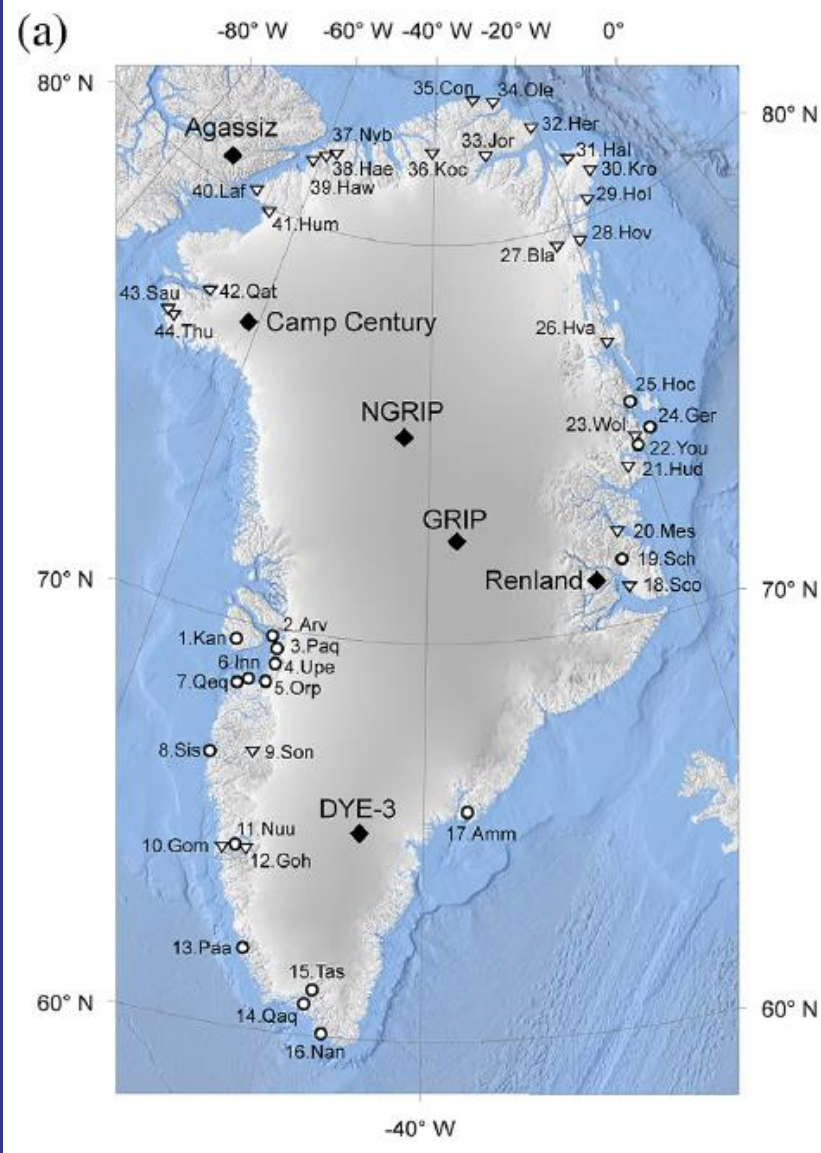
Model applications

Model tuning/calibration

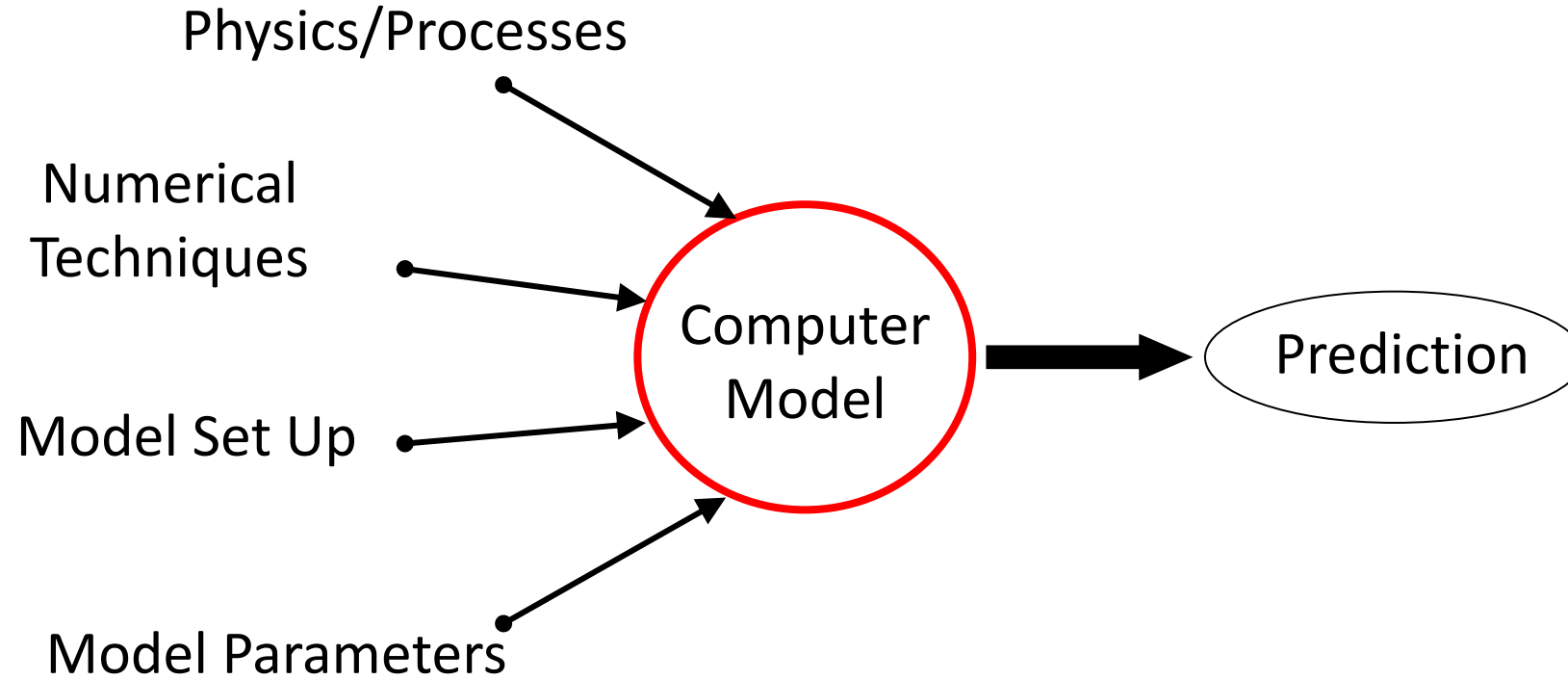
Interpreting data-model misfit

Model accuracy/uncertainty

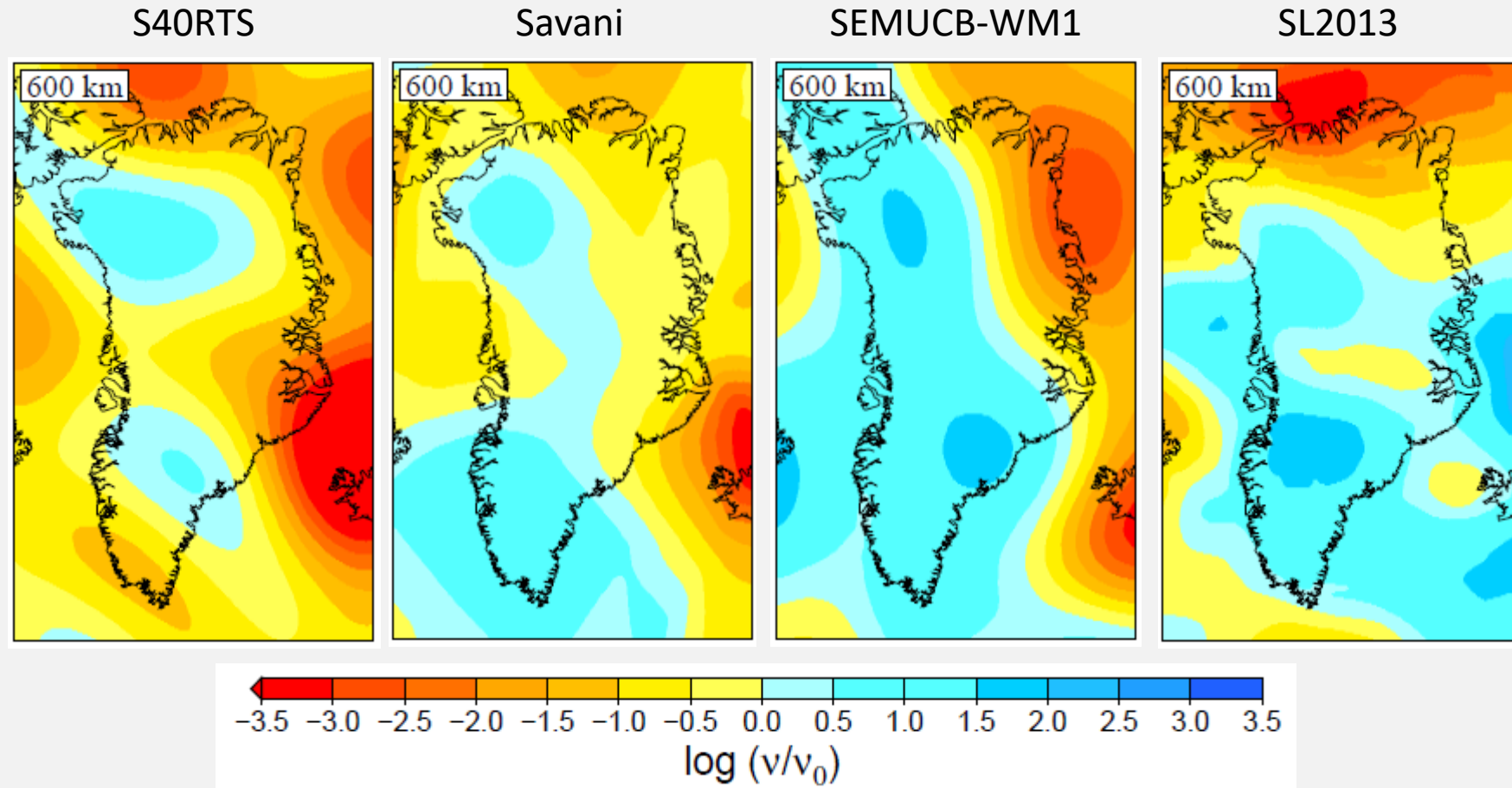
Interpreting Data-Model Misfits



General Elements of a Computer Model

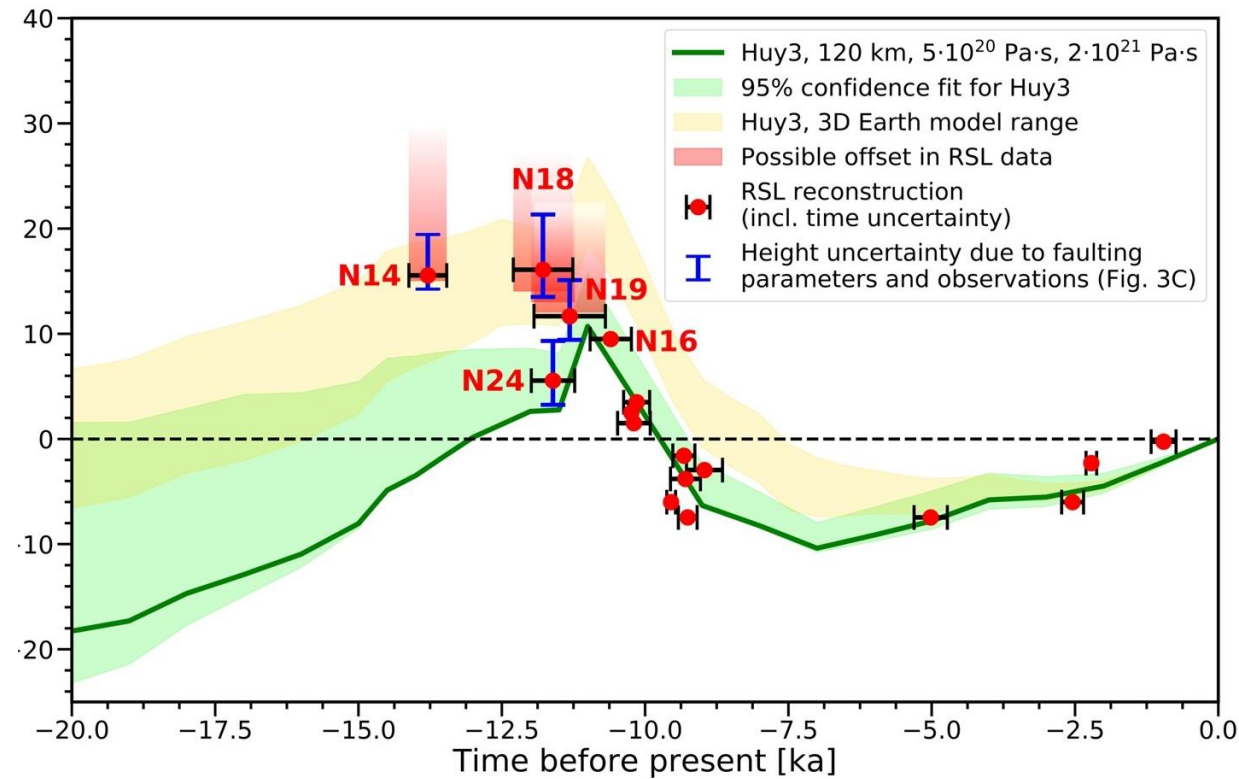
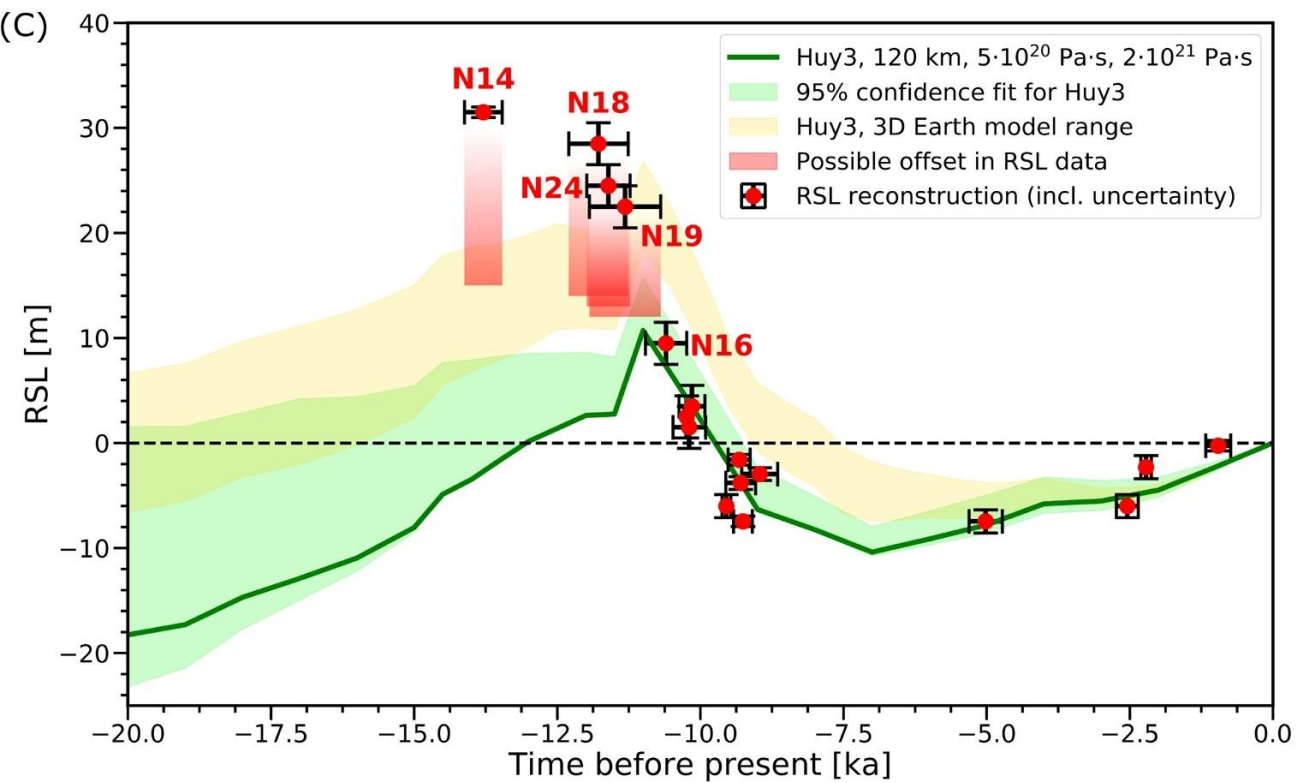


Model Set Up: Lateral Earth Structure



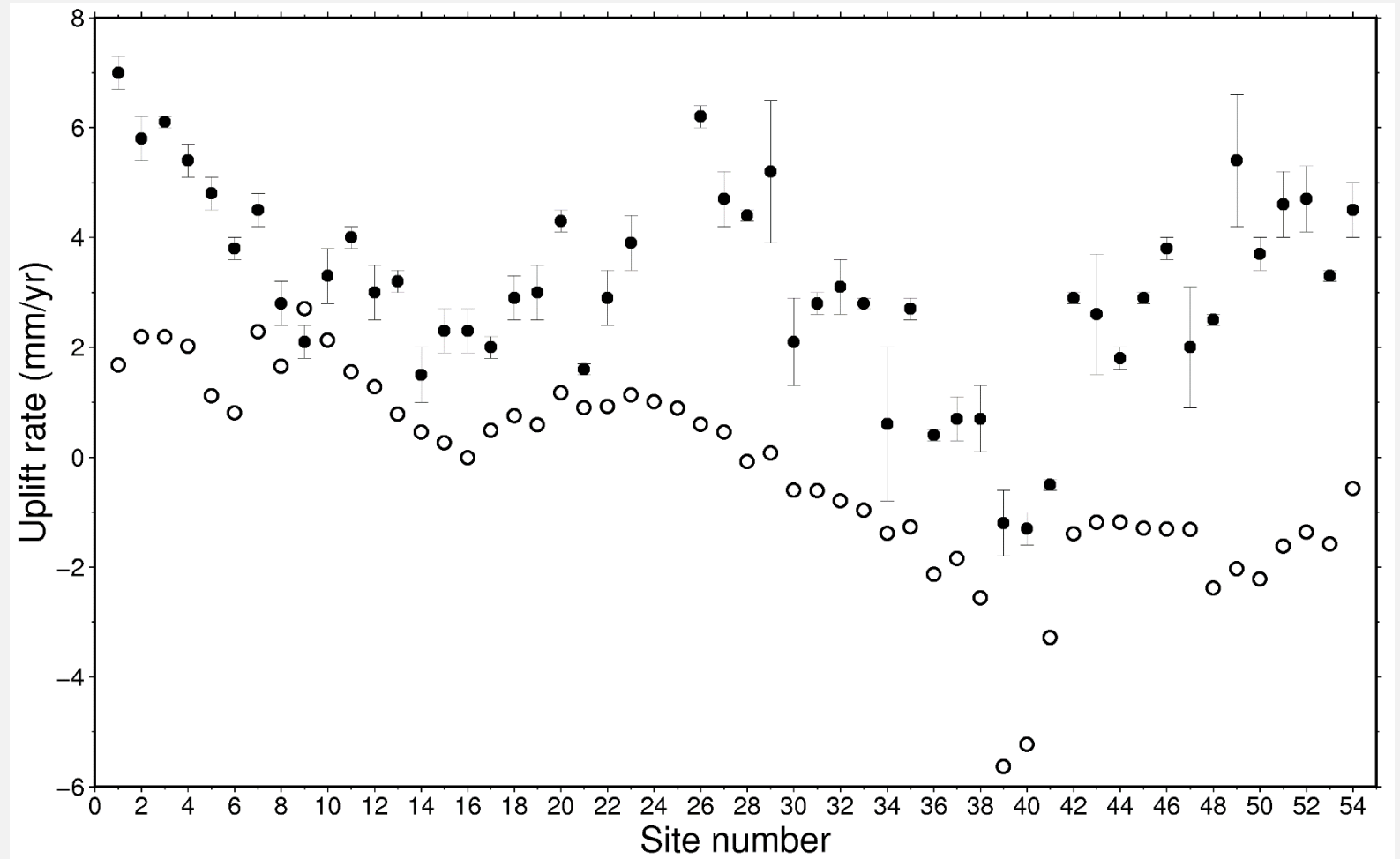
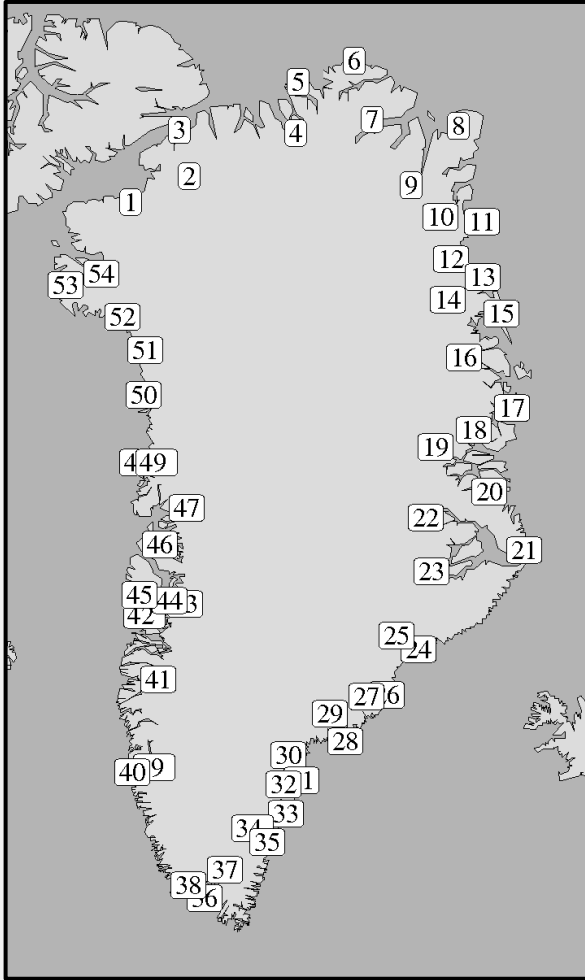
Milne et al. (2018)

Missing Processes



Steffen et al. (2020)

Compare Model Output to Other Data



Milne et al. (2018)

GIA Modelling and Model Applications

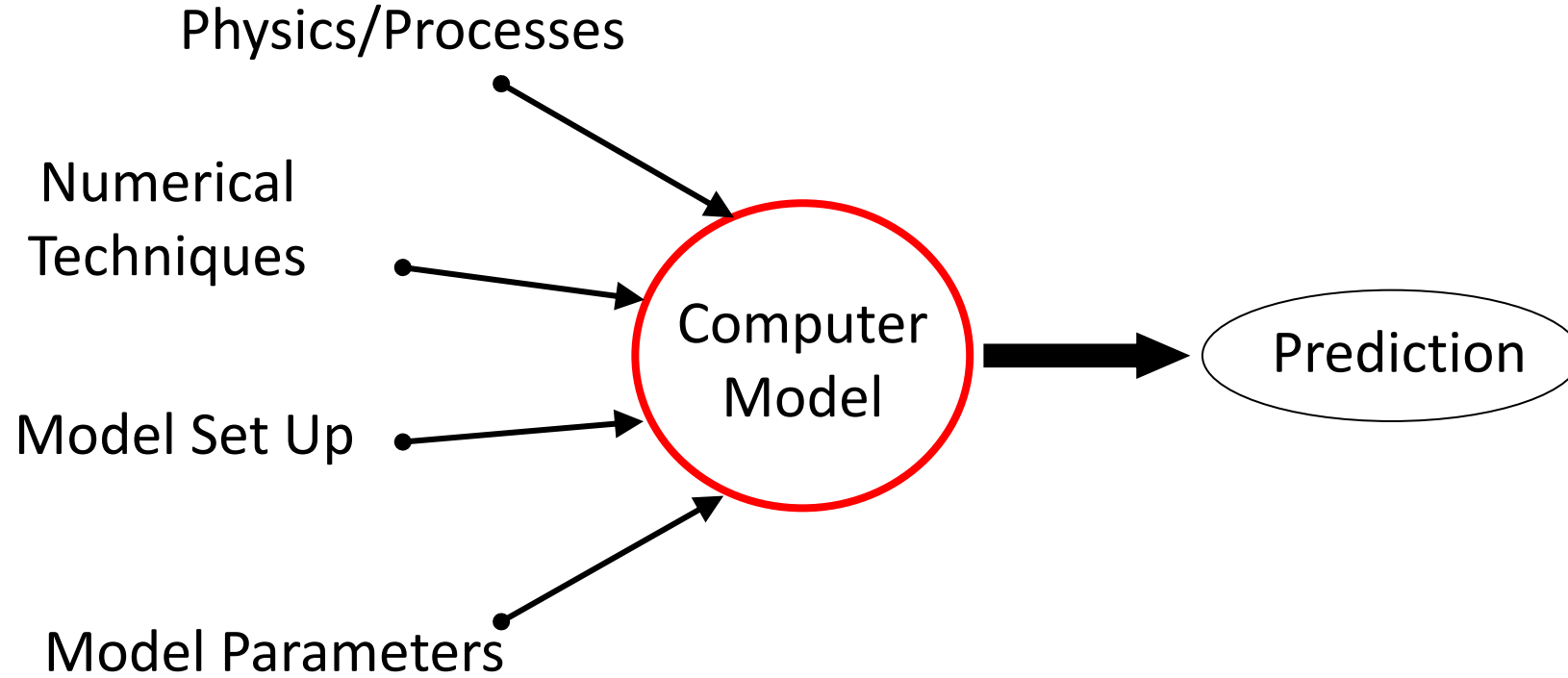
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Key Elements of a Computer Model



Model Accuracy/Uncertainty

Uncertainty in model input parameters defines “parametric uncertainty” of model.

Missing model physics, simplifications/errors in model setup, numerical errors lead to “structural error”.

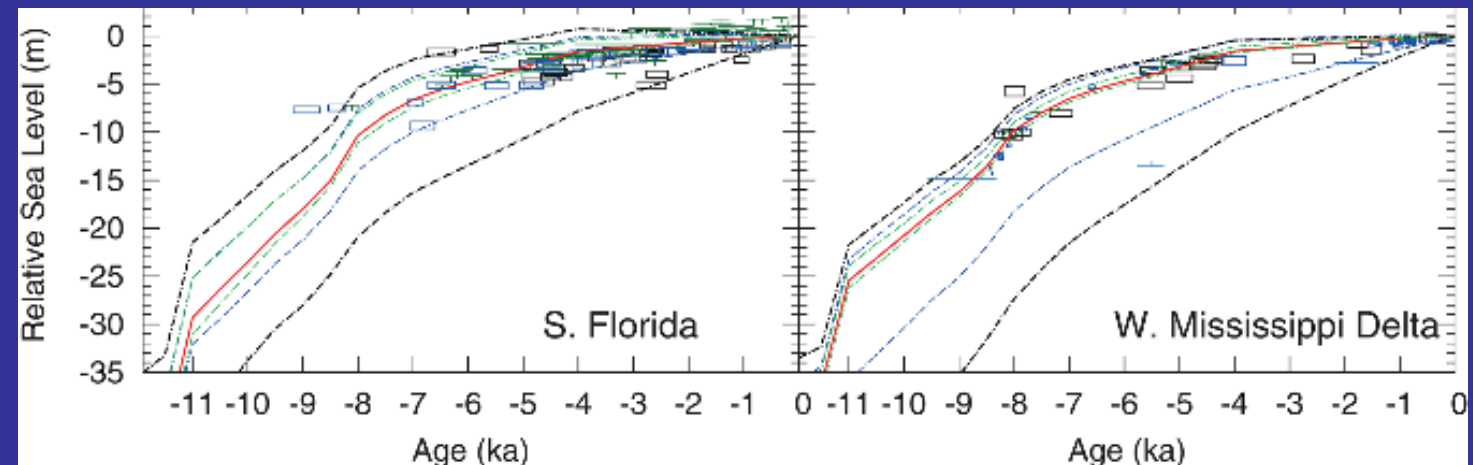
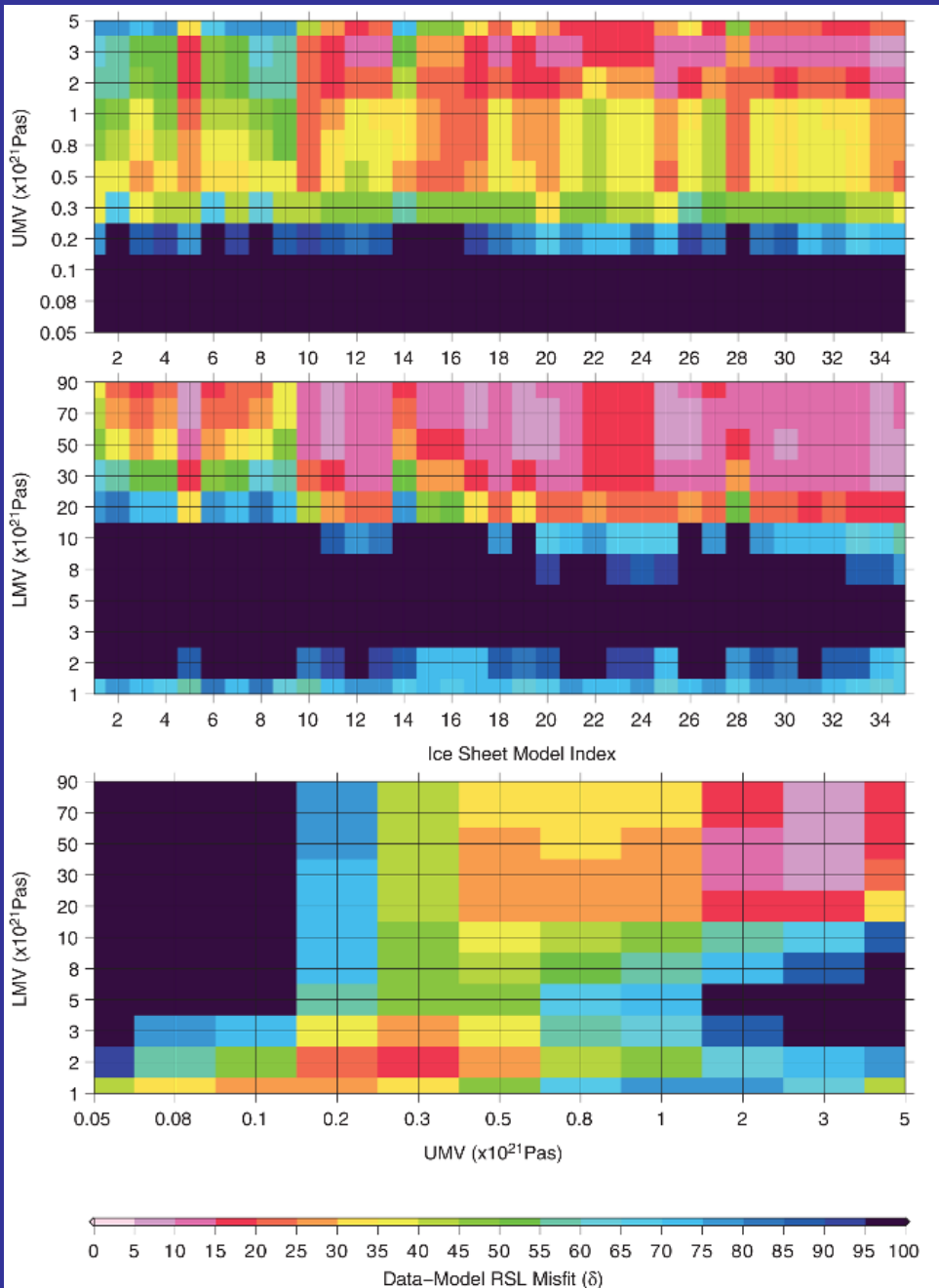
Both should be considered when using a model to make predictions that will be used in other applications.

Parametric Uncertainty

Need many model runs to accurately determine parametric uncertainty.

Different methods can be applied and compared.

Love et al. (2016)



Structural Error

One source is the common assumption of 1D spherically symmetric Earth structure.

There are many other sources of structural uncertainty (e.g., assumption of a given rheological model).

This aspect of model uncertainty is the most difficult to identify and quantify.

Data-Model Misfit: General Points

- The accuracy of a model is assessed by its ability to simulate observations.
- If model is unable to fit observations regardless of parameter values, this indicates that model is inaccurate (e.g., important processes are missing, model set up is inaccurate, numerical methods are inaccurate).
- Important note: even if a model can fit the observations, this is not a guarantee that it is accurate!